

1 Solution — GIAM 1.7

Problem 7

1.1 Problem

There is a relation known as “has color” which goes from the set

$$F = \{\text{orange, cherry, pumpkin, banana}\}$$

to the set

$$C = \{\text{orange, red, green, yellow}\}.$$

What pairs are in “has color”?

1.2 Answer

Pairing each fruit with its (ripe, typical) color:

has color = { (orange, orange), (cherry, red), (pumpkin, orange), (banana, yellow) }.

1.3 Reading Off the Pairs

The relation goes *from* F *to* C , so each pair is (fruit, color) with the fruit first:

fruit ($\in F$)	its color ($\in C$)	pair
orange	orange	(orange, orange)
cherry	red	(cherry, red)
pumpkin	orange	(pumpkin, orange)
banana	yellow	(banana, yellow)

Table 1.1

"has color" goes from fruits F to colors C . Draw an arrow from each fruit to the color it has.

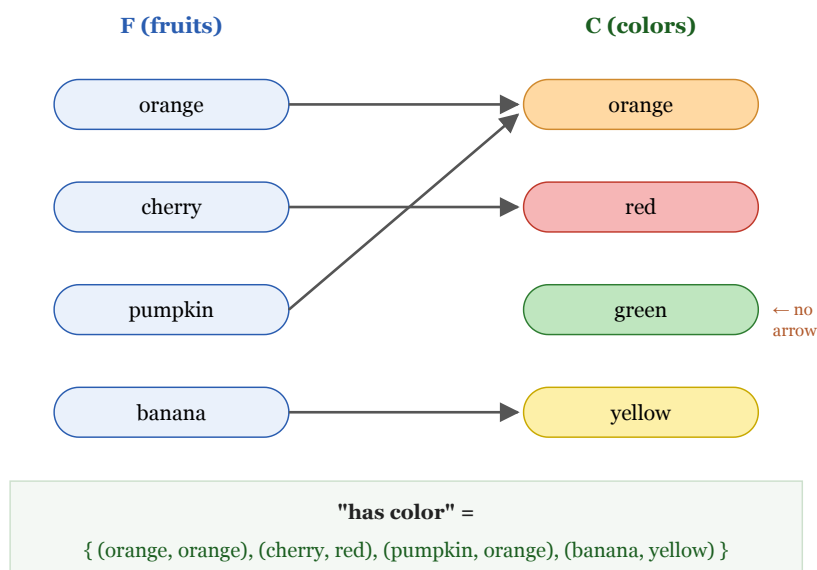


Figure 1.1: A bipartite arrow diagram from the fruit set F on the left to the color set C on the right, with the color nodes tinted their actual colors. Arrows: orange the fruit to orange the color, cherry to red, pumpkin to orange, banana to yellow. The color green on the right receives no arrow — no fruit in F is green. The pair-set is $\{ (orange, orange), (cherry, red), (pumpkin, orange), (banana, yellow) \}$.

1.4 Remark — “orange” Lives in Both Sets

The word *orange* appears in F (the fruit) **and** in C (the color). The pair **(orange, orange)** is therefore not a typo or a trivial “reflexive” pairing — its first coordinate is the *fruit* orange and its second is the *color* orange, two different kinds of thing that happen to share a name. Because a relation from F to C is a subset of the Cartesian product $F \times C$, the two coordinates come from different sets and are read in different roles; the coincidence of names is harmless. This is a small caution the problem builds in on purpose: in a relation *between two sets*, position (first vs. second coordinate) is what carries meaning, not the label alone.

1.5 Remark — Not Every Color Is Used (Not Onto)

The color **green** is in C but appears in *no* pair — none of these four fruits is green (in ripe form). So the **range** of the relation, **{orange, red, yellow}**, is a *proper subset* of the

codomain C . In the vocabulary of Problem 2, the relation is **not onto** C : there is a codomain element nothing maps to. Meanwhile *orange* (the color) is hit **twice** — by both the orange and the pumpkin — so if we view “has color” as a function it is **not one-to-one** either. Two fruits, one color: the many-to-one collapse means “has color” could not be inverted (Problem 6) — knowing something “has color orange” would not tell you whether it is an orange or a pumpkin.

1.6 Remark — This Relation Depends on Real-World Judgment

Unlike the arithmetic relations of the earlier problems ($|$, $\leq$, betweenness), “has color” is not settled by a formula — it depends on facts about the world, and those facts are fuzzy. A banana is *yellow* when ripe but *green* when unripe; a pumpkin is *orange* but some varieties are *green*; a cherry is usually *red* but can be nearly black or yellow. The answer above takes the *prototypical ripe color* of each fruit, which is surely the intended reading, but one could defensibly add pairs like **(banana, green)** or **(pumpkin, green)**. This is exactly the modelling-assumption theme that ran through Chapter 1: before listing the pairs, one must decide what “has color” is going to *mean*. State the assumption (here: ripe, typical color) and the relation becomes well-defined.

1.7 Remark — Why Relations Beat Functions for “has color”

Precisely because a single object can have several colors, “has color” is most honestly modelled as a **relation**, not a function. A function would force each fruit to exactly one color and could not record that a banana is *both* yellow and (when unripe) green. A relation carries no such restriction: it may contain **(banana, yellow)** and **(banana, green)** simultaneously. This is the deeper reason the section introduces relations as more general than functions — many natural real-world associations (a person “speaks” a language, a book “is about” a topic, a fruit “has” a color) are genuinely many-to-many, and only the relation framework captures them. With the ripe-color convention we happen to land on a function here, but the framework does not *require* it.